DESKLESS

Inspection reports, without the desk.

Walk the place, photograph each issue, say what is wrong. The phone writes the report, with no network.



Team techazsure | iQOO Hackathon 2026 Grand Finale | Productivity track | Bengaluru, 9 to 11 October 2026

02

Every place is walked and written up

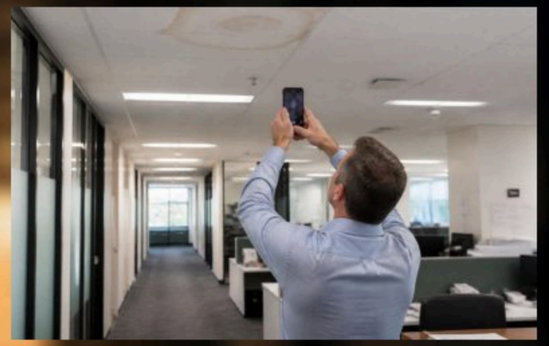
Someone walks every kind of place every day and has to write down what is wrong with it. Construction is one of them, not the centre.



Hospitals and clinics



Colleges and campuses



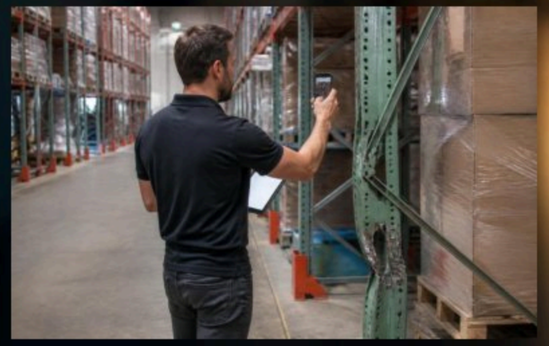
Government and company offices



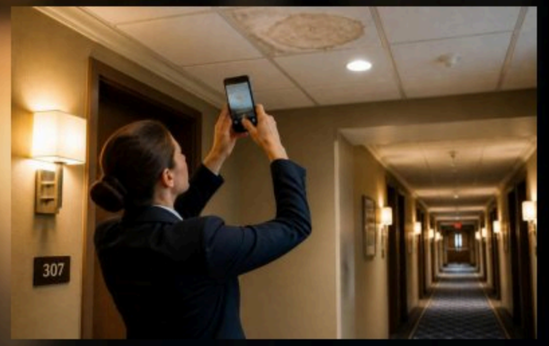
Apartment blocks and housing societies



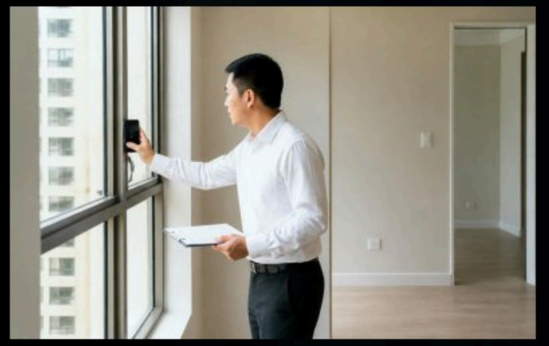
Malls and retail floors



Factories and warehouses



Hotels and hostels



Construction and handovers, one of many

03

The problem: the walk is one hour, the desk is another

What goes wrong today between seeing an issue and fixing it.



On the walk: issues are found in wards, corridors, basements and plant rooms, often with no network.

ANOTHER HOUR

at a desk, hours later, from memory and loose photos

Written up from memory
The write-up happens at a desk hours later, from memory and loose photos.

Photos with no context
They sit in WhatsApp groups with no location, owner or deadline attached.

Typed English on a small screen
Forms want English; people think and speak in Hindi, Kannada, Tamil or a mix.

Nobody checks the photo
No one asks whether the picture really shows what the note says.

The fixer hears last
The person who has to fix it hears last; management sees nothing until the report is typed.

Two hours of work for every hour of walking, and the record is still incomplete

04

Why it is needed, and how much

Scale of the problem in India, with the sources named.

USD 15.5 BN

India's integrated facility management market in 2025, heading to USD 24.9 bn by 2031 at about 8.3% a year (Mordor Intelligence)

~69,000

Hospitals in India (Statista estimate), where rounds, medication audits and expiry checks are daily routine

47,800+

Universities and colleges in India: 1,213 and 46,624 (AISHE 2022-23), each inspected for safety, hygiene and infrastructure

EVERY DAY

Someone in each of these places walks, looks and writes; that report is the only record. The cost is paid in hours at the desk, issues never written down, and repairs that start late

05

Existing solutions and their flaws

What places use today, and what each of them misses.

WHAT PLACES USE TODAY	WHAT EACH MISSES
Paper registers and WhatsApp groups free and familiar	✕ Nothing is structured, tracked or searchable ✕ The photo has no location, owner or deadline
Cloud inspection apps SafetyCulture, UpKeep, Fieldwire and alike	✕ Tap-through forms in typed English; about USD 24 per seat per month ✕ Offline mode only edits forms already on the device; reports and notifications wait for the network
Cloud AI add-ons voice notes and photos processed on a server	✕ The voice note and the photo leave the premises, which hospitals, government offices and housing societies cannot accept ✕ Nothing works where there is no network
All of them, for the people who use the place students, residents, patients, visitors	✕ None lets them report an issue in their own words and hear back when it is fixed

DESKLESS ✓ structured on the phone ✓ in your own language ✓ works with no network ✓ voice and photo stay on the device ✓ the public can report and hear back

06

Our solution: say it, the phone writes it

Photo plus voice in. Finished report out. All on the phone.

PHOTO + VOICE IN

- Photograph the issue or film a 20-second clip
- Say what is wrong in Hindi, English or a mix
- Position, compass, light and steps stamped

WHISPER HEARS

- Whisper small on the phone writes English text
- Long notes split at pauses
- Six Indian languages selectable

GEMMA 4 FILLS

- Issue, location, severity, owner, due date, type
- Says whether the photo matches the note
- Runs on the Adreno GPU, CPU fallback

SIGNED PDF OUT

- Issue map, sensor readings, per-owner action items
- Fingerprint signature from the Keystore
- Built in seconds, plus CSV and calendar

All four steps run in airplane mode
nothing is uploaded to make a report

with a network

The organisation's server
stores, ranks and routes the report

Every role's console
and the live linkage, as it happens

3 – 11 s

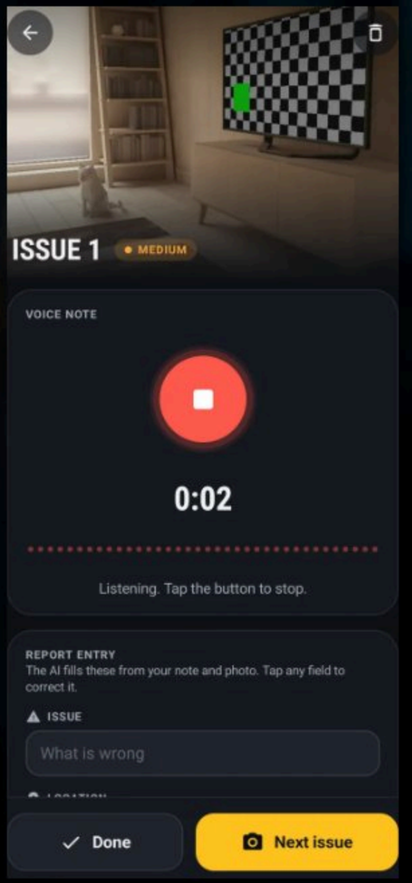
from the end of the recording to filled fields, measured on the iQOO 15

AIRPLANE MODE

capture, transcription, structuring and the PDF, end to end

HINDI · ENGLISH · MIX

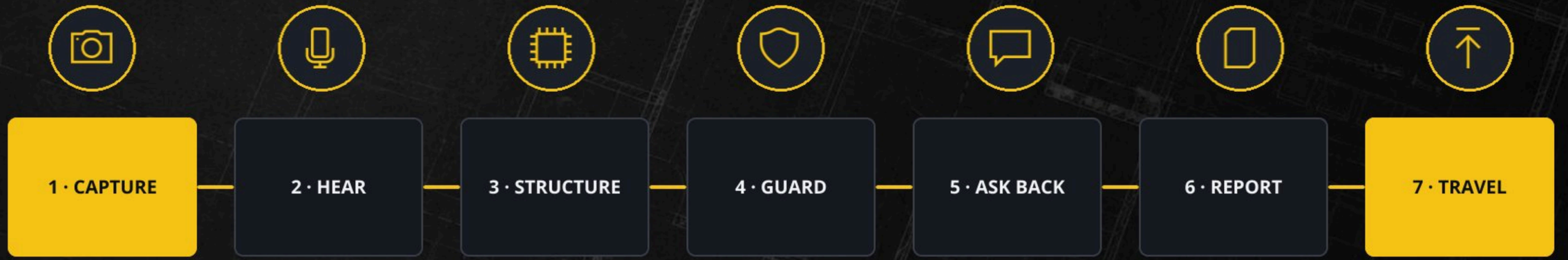
plus Kannada, Tamil, Telugu, Marathi and Bengali



07

How it works: from a voice note to a repair

Seven steps. The first five happen entirely on the phone; nothing is uploaded to make a report.



1 · CAPTURE

Photo or 20-second clip plus a voice note. Position, compass, light and step count stamped by the phone.

2 · HEAR

Whisper small on the phone turns Hindi, English or mixed speech into English text. Long notes split at pauses.

3 · STRUCTURE

Gemma 4 E2B on the phone's GPU reads the note and the photo, fills the five fields and judges whether the photo matches.

4 · GUARD

Plain code checks every value against what was said or seen. Anything else is held and shown, never written silently.

5 · ASK BACK

If where, who or by when is missing, the phone asks one question aloud and listens for the answer.

6 · REPORT

A signed PDF plus CSV, calendar and per-owner task list, and an AI review of the whole report.

7 · TRAVEL

To the server over HTTPS, phone to phone sealed when there is no network, or to a laptop through iQOO Office Kit.

08

Inside the phone: the AI that stays on the device

Technical depth on the on-device models and the rules around them.

ON THE PHONE · NOTHING UPLOADED

Whisper small · speech to English
via sherpa-onnx with Silero VAD; Hindi + English by default

Hindi · English · Kannada · Tamil · Telugu · Marathi · Bengali

Gemma 4 E2B · 2.6 GB · LiteRT-LM
Adreno GPU through OpenCL, CPU fallback; photo and text in, strict JSON out

3 – 11 s
from the end of the recording to filled fields, measured on the iQOO 15 with Gemma on the GPU

Models fetched by the app over Wi-Fi (2.96 GB, checksummed) or imported from a folder on a phone with no internet

The app has no internet permission for the report itself; the finished report goes to the organisation's own server when there is a network

THE RULE

THE MODEL
words things, fills forms

→

PLAIN CODE
decides and checks

Every field carries its source

heard · seen · phone · known · asked · told · judged · typed

GUARD AND MEMORY

A location, owner or deadline is written only if its words were said or seen. Otherwise it is held and shown, never written silently.

Kept out: owner "Suresh" was not in what you said
Use it · shown, not written

A phone-side memory learns owners and places from corrected fields; it is matched in code and never put in the prompt.

ASK, TELL, MEASURE

Ask or tell

"Mark issue 3 high, give it to Suresh, due Friday"

planned by code → applied only on Do it · Put it back
issue numbers come from the words, never from the model

Scorecard, measured on the phone

24 / 25 fields right
0 made up

20 fixed notes, spoken by the phone's own voice, heard by Whisper

The phone asks back

"Who should fix it, and by when?"

"Suresh the plumber, by Friday"

only the fields asked about are filled, tagged ASKED

09

The phone as an instrument

Sensors used as evidence, stamped without taps.

STAMPED WITHOUT TAPS

example readings

COMPASS

NE 42° faced

CAMERA AIMED

Overhead

LIGHT

320 lux

STEPS SINCE LAST ISSUE

48 steps

POSITION

satellite fix, drawn with no map tiles

On the map

PLACE TAG

Ward 3 NFC

INSTRUMENTS, ONE TAP EACH

LEVEL AND PLUMB

0.6° 10 mm/m

NOISE

72 dB

VIBRATION

1.2 m/s² 25 Hz

- NFC tags name the place once
- Lift to talk via the proximity sensor
- Torch runs off the light sensor
- Fingerprint sign-off, ECDSA P-256 in the Keystore

Readings go to the model with the note and into the PDF as a guide, marked as not calibrated.



When there is no network

Offline-first paths for the report.



all on the phone · no signal needed

AIRPLANE MODE

Capture, transcription, structuring and the PDF all happen on the phone. Nothing waits for a signal: a ward, a basement or a plant room is no different from the office.



PHONE TO PHONE

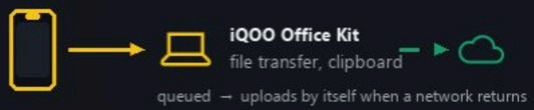
Nearby Connections over Bluetooth and Wi-Fi Direct. Every copy is sealed for one phone; relay phones carry what they cannot read; a signed receipt travels back.

ECDH P-256 · HKDF · AES-256-GCM



WALKIE-TALKIE AND HAND-OVER

Hold to talk: the clip rides the same sealed path. A user's entry with no network is handed to a nearby staff phone and posted when that phone finds a network.



OFFICE KIT AND UPLOAD

The PDF goes to the laptop by file transfer and the action items by the shared clipboard. Everything queued goes by itself the moment a network returns.

From report to repair: the server and the consoles

What happens once a report reaches the organisation.

ROLES AND WORKFLOW

Admin · Managing director · Manager · Inspector · Technical

Sign-up with an emailed code · approved by the level above · a unique DeskLess ID

OPEN (raised) → ASSIGNED (due date) → FIXED (repair photo, compared) → CLOSED (or sent back)

sent back

THE ASSISTANT

ranks every issue 0 to 100

Routine | Soon | Act now

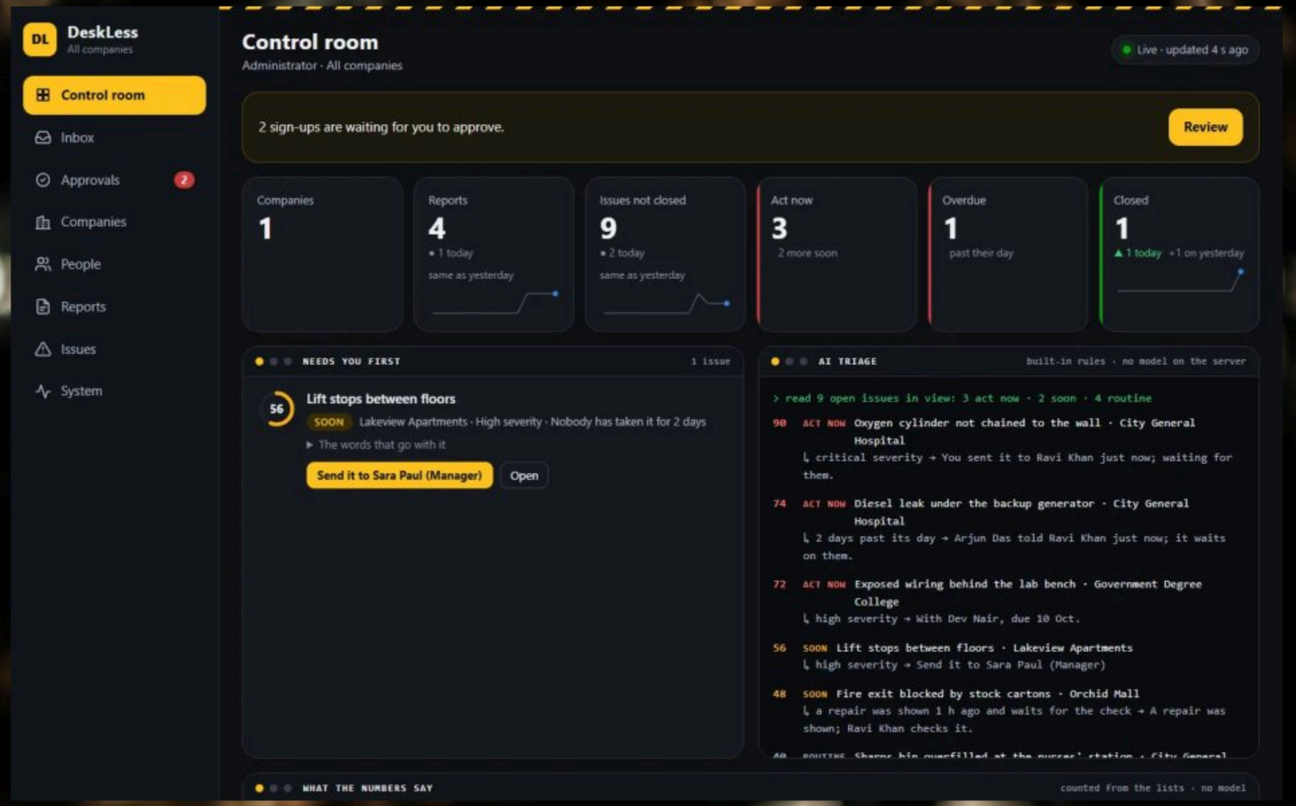
from: severity · deadline · hours untaken · danger words · photo mismatch · repeats at one place

- Names up to three people who could take it, with reasons
- Proposes the next step as a button; done on the second press
- Its own rounds chase untaken and late work by email

CONSOLES AND THE LIVE LINKAGE

Console per role · Inbox · Numbers in plain sentences · CSV export

Every phone, person, place and service drawn as nodes in 3D; each report, hand-off and answer flies between them as it happens.



The people who use the place close the loop

The DeskLess User edition and My desk.

DESKLESS USER · THE PUBLIC'S SIDE

Students, residents, patients and visitors pick the place and say what is wrong.

"Water leaking on the stairs near the lift"

Water leaking on stairs near lift
Where: stairs near the lift · Medium · Water and plumbing
read back aloud, then listening

"Make it high. Send it."

On the inspector's desk. The answer comes back to this phone and is read aloud.

MY DESK · SAID ALOUD, ANSWERED BY VOICE

Two need you first: the lift stops between floors at Lakeview...

"Give the first one to Arjun by Monday"

Give job 1 to Arjun Das, due 12 Oct · Do it

"I fixed it" opens the camera

~ 10 s
from a user's phone to the manager's desk

1.5 – 1.8 s
an instruction understood on the phone



Architecture

Three parts. The phone makes the report with no help; the server stores, ranks and routes; the web shows the work and lets each role act on it.

THE PHONE · makes the report

- **Capture** — photo, clip, voice, sensors
- **Whisper small** — speech to English, 6 languages
- **Gemma 4 E2B on GPU** — fills fields, checks the photo
- **Guard + memory** — code decides, model words
- **Sensors + NFC** — level, noise, light, place tags
- **Signed PDF pack** — Keystore ECDSA, CSV, ICS
- **Sealed relay** — Nearby, AES-256-GCM, receipts
- **Outbox** — uploads when online

no network needed for any of this

Kotlin · Jetpack Compose · sherpa-onnx · LiteRT-LM · CameraX · Nearby Connections

iQOO 15: 3 to 11 s from the end of the voice note to filled fields

Two editions from one code base: DeskLess (staff) and DeskLess User (the public)

Phone to phone with no network
Bluetooth, Wi-Fi Direct; sealed for one phone, relayed, receipted

iQOO Office Kit: PDF to the laptop by file transfer, action items by clipboard

HTTPS → upload

long poll ← to desk

THE SERVER · stores, ranks, routes

- **nginx** — under a path of the existing site, HTTPS
- **FastAPI + SQLAlchemy** — accounts, roles, reports, issues, workflow
- **PostgreSQL**
- **Assistant** — rank, route, next step
- **Writer, optional** — Gemma 3 1B, llama.cpp
- **Autopilot rounds** — chases late work
- **Mail** — SMTP, codes, hand-offs
- **Metrics + pulses** — requests, health, every hand-off for the live linkage

Python 3.12 · PostgreSQL · Docker Compose · deploy/setup.sh

Email to people
sign-up codes, issues passed on, the assistant's rounds

110 Android unit tests · 99 server tests · on every push

JSON → long poll

actions ← assign, close

THE WEB · shows and acts

- **Home page + sign-up** — film, deck, screens, the way in
- **A console per role** — MD, manager, inspector, technical
- **Inbox · issues · reports** — search, CSV export, live refresh
- **Report types + public door** — a company's own types
- **Live linkage** — every node and hand-off in 3D

Plain HTML, CSS and JS · no build step

One command
docker compose up builds both APKs and runs it all

GitHub Actions builds and tests the same tree

Tech stack

App
one APK, two editions

Kotlin 2.2 · Jetpack Compose · CameraX · minSdk 24 · staff + user editions from one code base

Speech and voice
on the phone

Whisper small · sherpa-onnx 1.13 · Silero VAD · Android text-to-speech · Hindi · English · Kannada · Tamil · Telugu · Marathi · Bengali

Language and vision model
on the phone, optional on the server

Gemma 4 E2B (.litertlm) · LiteRT-LM 0.18 · OpenCL GPU, CPU fallback · Gemma 3 1B via llama.cpp on the server (optional)

Sensors, security, phone to phone
on the phone

SensorManager · NFC · BiometricPrompt · Keystore ECDSA P-256 · AES-256-GCM sealing · Nearby Connections (P2P_STAR) · ZXing QR pairing · HMAC proof

Server, web and ops
one container beside the website

Python 3.12 · FastAPI · SQLAlchemy · PostgreSQL · scrypt · HS256 tokens · plain JS + Canvas 2D · Docker Compose · Caddy · nginx · GitHub Actions · 110 Android + 99 server tests

runs on the phone · runs on the server

What makes DeskLess different

Uniqueness and innovation.

01

AI on the phone, in Indian languages

Nothing is uploaded to make a report. Hindi, English and four more languages, heard and written on the device.

02

Evidence over eloquence

The model words, plain code decides. Every field says where it came from; what was not said is held, not invented.

03

A sealed relay between phones

Reports and voice messages cross phones that cannot read them, with signed receipts back to the sender.

04

The phone as an instrument

Compass, level, noise, vibration, light, NFC tags and a fingerprint signature, stamped as evidence.

05

Both sides of the place

The inspector's rounds and the public's voice meet on one desk that talks back, in a hospital, a college or a housing society alike.

06

An assistant that explains itself

Ranks, routes and chases with its reasons on screen, and a 3D live view of the whole system as it runs.

Feasibility: it runs today

Proof that it is built, with measured numbers.

3 – 11 s
voice note to filled fields on the iQOO 15

110 + 99
Android and server tests, passing on every push

2 phones
iQOO 15 on the Adreno GPU, and a Pixel 9a

LIVE
shaikalthaf.in/competitions/iqoo, sign-up to consoles

PROOF

- Built in the 48-hour finale: a native Android app plus a server, public on GitHub
- Deployed with one script beside an existing website, sign-up to live linkage
- One command, docker compose up, builds both APKs, runs every test, serves it all
- Models fetched by the app itself and checksummed; nothing installed by hand



Scalability

How it grows from one team to many organisations.

GROWS WITHOUT A REWRITE

the same app and server, from one team to many tenants



Any kind of place
Twelve report types built in, plus a company's own, pushed to phones at check-in

Multi-tenant by design
Each organisation has its own roles, people, report types and public door

AI on phones already in hand
Inference scales with the phones people own; the server only stores, ranks and routes

No per-report cloud cost
Models fetched once per phone (2.96 GB); nothing is paid per report

Stateless server
FastAPI behind nginx with PostgreSQL: add instances behind the same path; long polls hold no database connection

Six Indian languages today
More by swapping the speech model file; no change to the app

Practicality and adoption

Why an organisation can start on Monday.



START ON MONDAY

1. **A managing director signs up**
   name, email, a code by email
2. **Invites a manager with a code**
   eight characters, ABCD-EFGH
3. **Inspectors and technicians join**
   under their manager, approved above

Nothing to buy
A mid-range Android with 8 GB runs it; the demo runs on a Pixel 9a

Nothing to learn
Walk, photograph, speak; the phone asks when something is missing

FITS HOW PLACES ALREADY WORK

- The PDF, CSV, calendar file and per-owner task list go wherever reports go today, including by Office Kit to a laptop
- Privacy by construction: voice and photo are processed on the phone; identity numbers are encrypted and shown masked
- Every report is signed on the phone and the signature is re-checked wherever it lands
- It keeps working where the network does not: wards, basements, plant rooms, new sites

Where it goes next

Roadmap, three horizons.

NOW — + 3 MONTHS — + 12 MONTHS

NEXT 3 MONTHS

- Pilots in one hospital, one college and one housing society
- Multi-hop relay and background relaying
- A push service so desks announce while the app is closed

NEXT 12 MONTHS

- iOS edition
- More Indian languages
- Office Kit integration when an SDK is published
- Analytics across organisations, with consent

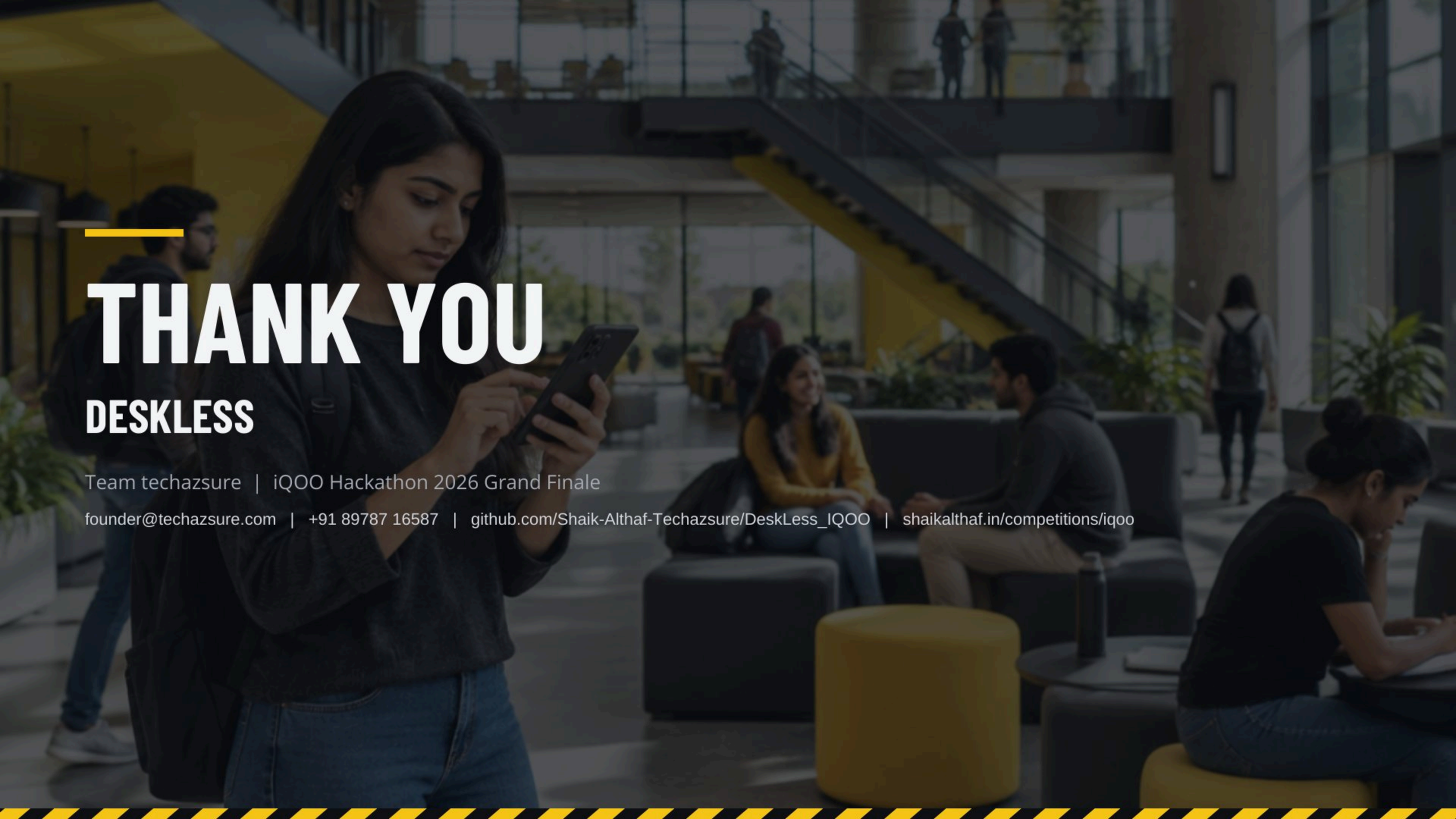
BUSINESS MODEL

FREE small teams
the app, the on-phone AI, the signed report

PER PLACE subscription
consoles, assistant, server and report types

ENTERPRISE self-hosted licence
its data stays on its own server, with support

The on-phone AI stays free: it costs the organisation nothing to run. Full canvas in the companion document.



THANK YOU

DESKLESS

Team techazsure | iQOO Hackathon 2026 Grand Finale

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